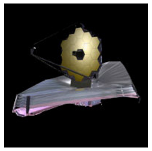

6061 - A Systematic Study of the 3.3 - 3.5 micron PAH Features at z~0 with Archival NIRSpec Observations

Cycle: 3, Proposal Category: AR

INVESTIGATORS

Name	***Institution***
Dr. Karin Marie Sandstrom (PI)	**University of California - San Diego**
Dr. Thomas Lai (CoI)	California Institute of Technology
Dr. Ryan Chown (CoI)	The Ohio State University
Dr. Jeremy Chastenet (CoI) (ESA Member)	Ghent University
I-Da Chiang (CoI)	Academia Sinica Institute of Astronomy and Astrophysics
Dr. Thomas Williams (CoI) (ESA Member)	University of Oxford

OBSERVATIONS

ABSTRACT

We propose an archival study of all publicly available NIRSpec IFU observations that cover the 3.3-3.5 micron PAH features in z~0 extragalactic and Milky Way observations of the ISM. Uniformly analyzing this growing compilation is an efficient way to build a comprehensive sample of 3.3-3.5 micron PAH measurements, spanning a wide enough range of ISM and galactic environment to unlock the diagnostic potential of these emission features. As the PAH features detectable out to the highest redshifts and the focus of wide-field NIRCam medium band mapping capabilities at z=0, the 3.3-3.5 micron PAHs are some of the most valuable tracers we have of the ISM in galaxies, but they are also the most poorly characterized, since they were outside the spectroscopic range of Spitzer. Using our archival compilation, we will investigate how the PAH populations change with environment and establish the z=0 baseline for comparison with high redshift galaxies.

OBSERVING DESCRIPTION